PROPOSITIONAL LOGIC

Logical Proposition
↳ short statement which can either be true or false
(ex. $x = 10$, $y < 1$, $a \neq p$)

Tautology → Something that is always true
Contradiction → Something that is always false

$P \wedge Q$ (P **and** Q are both true)

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

conjunction

$P \vee Q$ (P **or** Q: either or both is true)

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

disjunction

$\neg P$ ("not" P, negation)

P	$\neg P$
T	F
F	T

$P \Rightarrow Q$ (P implies Q or if P then Q)

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Logical Proposition

$P \Leftrightarrow Q$ (P "if and only if" Q)

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

PROOF

1. To show 2 propositions are not logically equivalent it is enough to find values which aren't true

Theorem 1.3.1

$P \wedge P \Leftrightarrow P$

$P \vee P \Leftrightarrow P$

$P \wedge Q \Leftrightarrow Q \wedge P$

$P \vee Q \Leftrightarrow Q \vee P$

$P \wedge (Q \wedge R) \Leftrightarrow (P \wedge Q) \wedge R$

$P \vee (Q \vee R) \Leftrightarrow (P \vee Q) \vee R$

$P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$

$P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$

all of these are tautologies

De Morgan's Laws

Theorem 1.3.2

$P \vee \neg P \Leftrightarrow T$

$P \wedge \neg P \Leftrightarrow F$

$P \Leftrightarrow \neg(\neg P)$

$\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$

$\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$

$\neg T \Leftrightarrow F$

$\neg F \Leftrightarrow T$

Theorem 1.3.3

$P \vee F \Leftrightarrow P$

$P \wedge T \Leftrightarrow P$

$P \wedge F \Leftrightarrow F$

$P \vee T \Leftrightarrow T$

Theorem 1.3.4

$(P \Rightarrow Q) \Leftrightarrow (\neg P \vee Q)$

$(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$

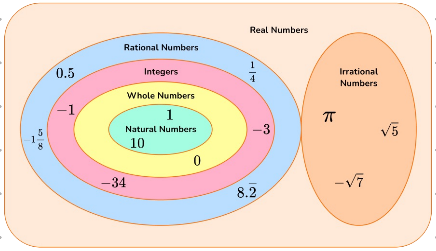
$(P \Leftrightarrow Q) \Leftrightarrow (P \Rightarrow Q) \wedge (Q \Rightarrow P)$

$P \Leftrightarrow (\neg P \Rightarrow F)$

SETS AND FUNCTIONS

Sets

Set: collection of distinct elements, written as $A = \{a_1, a_2, a_3\}$



Belonging: $a \in A$ a belongs to A
$a \notin A$ a is not an element of A

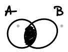
Equal Sets: 2 sets are only equal if all of their elements are equal

Subset vs proper Subset:

Subset ($\subseteq$) $A \subseteq B$
- every element of A is in B
- includes possibility of $A = B$

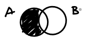
Proper Subset ($\subset$) $A \subset B$
- every element of A is in B
- $A \neq B$

Set Operations:



1) **Intersection**
$A \cap B = \{a \mid a \in A \cap a \in B\}$

2) **Union**
$A \cup B = \{a \mid a \in A \cup a \in B\}$



3) **Difference**
$A \setminus B = \{a \mid a \in A \cap a \notin B\}$

Cartesian Product

$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$

Functions

Function: $f: A \to B$ maps each $a \in A$ to a unique $b \in B$

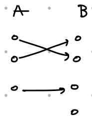
Domain: set A Codomain: set B

Identity Function: maps every element of domain to itself $id_A(x) = x \ \forall x \in A$
- Domain = Co-domain
- Bijective
- is its own inverse

Image: $Im(f) = \{f(x) \mid x \in A\}$
- subset of co-domain
- actual set of the values obtained by the function

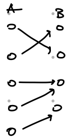
Types of Functions:

Injective: $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$
↳ every input → 1 output



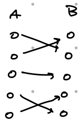
- B can't have many A's

Surjective: every $b \in B$ has at least 1 $a \in A$, such that $f(a) = B$



- Every B has some A
- Image = co-domain

Bijective: Both injective & surjective
A to B perfectly



Composite Function:
- $g \circ f(x) = g(f(x))$

Inverse Function:
- g is inverse of f if $g(f(a)) = a$ AND $f(g(b)) = b$
- an f can only have an inverse if it is bijective

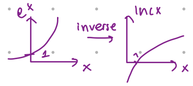
Useful Lemmas:

Lemma 2.2.2:
$f: A \to B$
f is bijective if and only if f has an f^{-1}

Lemma 2.3.1:
$f: A \to B \mid A, B \subseteq \mathbb{R}$
either $\forall a_1, a_2 \in A$
$a_1 < a_2 \Rightarrow f(a_1) < f(a_2)$
or $\forall a_1, a_2 \in A$
$a_1 < a_2 \Rightarrow f(a_1) > f(a_2)$
f is injective

Strictly monotone Functions
(1) A function where $a_1 < a_2 \Rightarrow f(a_1) < f(a_2)$ for All $a_1, a_2 \in A$ is strictly increasing
(2) A function where $a_1 < a_2 \Rightarrow f(a_1) > f(a_2)$ for All $a_1, a_2 \in A$ is strictly decreasing
- strictly monotone functions are injective!

ex. $x \mapsto e^x$ is strictly increasing



⇒ injective
⇒ has an inverse

Limiting co-Domain to obtain Inverse

$f: \mathbb{R} \to \mathbb{R}$ $g(x) = x^2$ → not bijective
↳ no inverse

$g: \mathbb{R} \to \mathbb{R}_{\geq 0}$ $g(x) = x^2$ → surjective
b.c $g(\mathbb{R}) = \mathbb{R}_{\geq 0}$ which is co-domain of g

$h: \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ $g(x) = x^2$ → bijective
↳ has inverse

Trigonometric Functions:

sine, cosine
- surjective
image of sin is $[-1, 1]$ which is also co-domain
- not injective:
multiple input values have the same output

tangent
$\{x \in \mathbb{R} \mid \cos(x) \neq 0\} \to \mathbb{R}$
where $\tan(x) = \frac{\sin(x)}{\cos(x)}$
- surjective
image = co-domain
- not injective
many outputs with the same input

Complex numbers and polynomials

Introduction to complex numbers

Complex Field ($\mathbb{C}$) = $\{a+bi \mid a, b \in \mathbb{R}\}$

↑ set of numbers that follows all properties and defines + & −

Properties:
- commutative
- associative
- distributive

Identities:
- symmetric
- transitive
- additive
- multiplicative

Neutral identities:
- additive
- multiplicative

Inverses:
- additive
- multiplicative

Components of a complex number:
$Re(z) \rightarrow$ real part $Im(z) \rightarrow$ imaginary part

Argand Diagram:
$x \rightarrow$ real axis $y \rightarrow$ imaginary axis
distance → modulus
Argument → θ formed with x-axis

Modulus:
$|z| = \sqrt{a^2+b^2}$

Argument

Arithmetic with complex numbers

Addition, Subtraction:

$(a+bi)+(c+di) = (a+c)+(b+d)i$
$(a+bi)-(c+di) = (a-c)+(b-d)i$

Multiplication

$(a+bi)(c+di) = ac+adi+bci-bd$

Division

$$\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{(c+di)(c-di)} = \frac{ac-adi+bci+bd}{c^2+d^2}$$

Complex conjugate

if $z = a+bi$ then its complex conjugate $\bar{z}$ is $\bar{z} = a-bi$

Property: $z\bar{z} = a^2+b^2 = |z|^2$

Polar and Euler form

Polar Form: $z = r(\cos\theta + i\sin\theta)$
Euler Form: $z = re^{i\theta}$

Rectangular → Polar
$r = \sqrt{a^2+b^2}$
$\theta = \arctan\left(\frac{b}{a}\right)$
$z = r(\cos\theta + i\sin\theta)$

Polar → Rectangular
$a = r\cos\theta$
$b = r\sin\theta$
$a+bi$

De moivre's theorem

For a complex number $z = re^{i\theta}$

$$z^n = r^n e^{in\theta}$$

Finding an n^{th} root:

$$z_k = r^{\frac{1}{n}} e^{\frac{i(\theta+2k\pi)}{n}} \quad k=0,1,\dots n-1$$

Roots of complex numbers

Roots of complex numbers: solutions to the equations $\omega^n = z$, where $n \in \mathbb{R}$ and $z = a+bi$

General Form of Roots:
Roots of $|z|e^{i\theta}$ are

$$w_k = \sqrt[n]{|z|}\, e^{i\left(\frac{\theta+2k\pi}{n}\right)} \quad k=0,1,\dots,n-1$$

Finding a n^{th} root:

General Method:

1) Compute modulus of the root (s):
$|w| = \sqrt[n]{|z|}$

2) Determine arguments
$\theta_k = \frac{\theta+2k\pi}{n}$ for $k=0,1,\dots,n-1$

3) Express the roots $w_k = |w|e^{i\theta_k}$

Ex

$z = 8i$ compute cubed roots

1) Express in polar form
$z = 0+8i$
$\arg(z) \rightarrow 8i$ lies on $\frac{\pi}{2}$ so
$\theta = \frac{\pi}{2}$
$z = 8e^{i\frac{\pi}{2}}$

2) compute modulus of roots
$|w| = \sqrt[3]{|z|} = 2$

3) Determine arguments

$\theta_1 = \frac{\frac{\pi}{2}}{3}$ $\theta_2 = \frac{2\pi+\frac{\pi}{2}}{3}$
$\theta_3 = \frac{4\pi+\frac{\pi}{2}}{3}$ $\theta_1 = \frac{\pi}{6}$ $\theta_2 = \frac{5\pi}{6}$
$\theta_3 = \frac{3\pi}{2}$

4) Express the roots:
$w_1 = 2e^{i\frac{\pi}{6}}$ $w_2 = 2e^{i\frac{5\pi}{6}}$
$w_3 = 2e^{i\frac{3\pi}{2}}$

Properties of polynomials

Polynomial Degree: Highest power of z, with a non-0 coefficient

Leading coefficient: Coefficient a_n of the term with the highest power of z

Polynomials of degree 2:

$D = b^2 - 4ac$:
$D > 0$: 2 Real Roots
$D = 0$: 1 Real (Repeated) Root
$D < 0$: 2 complex conjugate Roots

To find Roots:

$$z = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

Real Coefficent Polynomials:

1) Complex Conjugate Roots:
→ if polynomial is real, but has a complex root, it must have a 2nd conjugate root
→ $(a+bi)$ & $(a-bi)$

Binomials

- $z^n - w = 0$ is a binomial equation
- All roots are given by:

$$z_k = \sqrt[n]{|w|}\, e^{i\left(\frac{\arg(w)+2k\pi}{n}\right)}$$

$k = 0, 1, \dots, n-1$

Polynomials and Recursion

The Division Algorithm

The division Algorithm for Polynomials:

- For any polynomials $f(z)$ and $g(z)$ $(g(z) \neq 0)$, there exists unique polynomials $q(z)$ (quotient) and $r(z)$ the remainder such that:

$$f(z) = g(z) \cdot q(z) + r(z)$$

where $\deg(r(z)) < \deg(g(z))$

- if $r(z) = 0$, $g(z)$ is a factor of $f(z)$
- Dividing $f(z)$ by $(z-\lambda)$ helps check if λ is a root

Dividing Polynomials Systematically

$$\begin{array}{r|l} 2z^3+3z^2-5z-6 & z+2 \\ \hline -\underline{(2z^3+4z^2)} & 2z^2-z-3 \\ -z^2-5z & \\ -\underline{(-z^2-2z)} & \\ -3z-6 & \\ -\underline{(-3z-6)} & \\ 0 & \end{array}$$

Roots and Factorization

- **Root:** A complex number λ is a root of $p(z)$ if $p(\lambda)=0$
- **Multiplicity:** A root λ has multiplicity m if $(z-\lambda)^m$ divides $p(z)$, but $(z-\lambda)^{m+1}$ does not.
- **Factoring Polynomials**: Any polynomial can be factored into linear factors:

$$p(z) = a_n(z-\lambda_1)(z-\lambda_2)\cdots(z-\lambda_n)$$

where λ roots are real or complex

ex $p(z) = z^3-6z^2+11z-6$ Roots $= 1, 2, 3$

$p(z) = (z-1)(z-2)(z-3)$

Recursively defined functions

- **Recursive function**: defined by using the functions previous values

ex
$$n! = \begin{cases} 1 & \text{if } n=0 \\ n\cdot(n-1)! & \text{if } n>0 \end{cases}$$

- **Recursive algorithm**:

Base case: value for the smallest input

Recursive step: Define larger inputs in terms of smaller ones

Method:
1. Define base case
2. Recursive Formula
3. Solve Recursively or use iterative computation

ex

Calculate $5!$:

B.C. : $1! = 1$

Recursive step: $5! = 5\cdot 4! = 120$

Summation symbol

Recursive Summation

Recursive Symbol: $\sum_{k=1}^{n} k = \sum_{k=1}^{n-1} k + n$

Base Case: $\sum_{k=1}^{1} k = 1$

Explicit Formula for Summation:

$$\sum_{k=1}^{n} n = \frac{n(n+1)}{2}$$

SYSTEMS OF LINEAR EQUATIONS

STRUCTURE OF SYSTEMS OF LINEAR EQUATIONS

System of Linear Equations: A collection of multiple linear equations involving the same variables. The solution to a system is a set of values for the variables that satisfies all equations simultaneously.

Homogeneous vs. Inhomogeneous:

A homogenous system has all constants b equal to zero
An inhomogenous system includes at least one non-zero constant b

ex. $\begin{cases} 2x_1 + 3x_2 = 5 \\ 4x_1 - x_2 = 6 \end{cases}$ → This is an inhomogeneous, as there are non-zero solutions that satisfy it

Inhomogeneous

★ Rightmost column non-zero

Particular sltn: 1 specific sltn
$Ax = B$
General sltn:
$x = x_p + x_h$
where: x_p is a particular sltn to an inhomogeneous system
x_h is any sltn to the corresponding homogeneous system

Homogeneous

★ Rightmost column is 0s.

→ If w & u are solutions to a homogeneous system, then w + u is still a solution (on the line)

→ If w is a solution to a homogeneous system, then cw (where c is a scalar), it is still a solution (still on the line)

→ It can not have 0 solutions, bc 0 is always a solution

→ If a 0-row (free variable), ∞-many solutions

→ Solutions to homogeneous system create a vector space (meaning they can be added tg & scaled)

TRANSFORMING A SYSTEM OF LINEAR EQUATIONS

Elementary Row Operations

Row Interchange: Swap two rows.
Row Scaling: Multiply a row by a non-zero constant.
Row Replacement: Replace a row with the sum of itself and a multiple of another row.

Purpose of Row Operations

These operations are used to simplify systems while preserving the solution set, eventually leading to a form that's easier to solve.

Matrix Representation

Coefficient Matrix: Formed by the coefficients of the variables.
Augmented Matrix: Coefficient matrix with an added column for the constants.

ex $\begin{cases} 2x_1 + 3x_2 = 5 \\ 4x_1 - x_2 = 6 \end{cases} \Leftrightarrow \left[\begin{array}{cc|c} 2 & 3 & 5 \\ 4 & -1 & 6 \end{array}\right]$

COMPUTING ALL SOLUTIONS TO A SYSTEM OF EQUATIONS

Solution Types

Particular Solution: A specific solution that satisfies the inhomogeneous system.
General Solution: Expressed as the sum of a particular solution and the solution to the homogeneous system.
Existence and Consistency

A system has a solution if the rank of the coefficient matrix matches the rank of the augmented matrix. Otherwise, the system is inconsistent.

Relation of rank to the number of solutions

1) No Solutions
   - $\rho(\underline{\underline{I}}) > \rho(\underline{\underline{A}})$
2) 1 Solution
   - $\rho(\underline{\underline{I}}) = \rho(\underline{\underline{A}}) = n$
   - if n is # of variables
3) ∞ - many Solutions
   - $\rho(\underline{\underline{I}}) = \rho(\underline{\underline{A}}) < n$

Rank: # of pivots

THE REDUCED ROW ECHELON FORM OF A MATRIX

Definition of RREF

A matrix is in reduced row echelon form if:

1. Each leading entry (pivot) in a row is 1 and is to the right of the leading entries in rows above.
2. Each pivot is the only non-zero entry in its column.
3. Rows consisting entirely of zeros are at the bottom.

without this requirement, it is said that matrix is in row-echelon form.

ex $\begin{bmatrix} 2 & 3 & 5 \\ 4 & -1 & 6 \end{bmatrix} \Leftrightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$

Number of parameters based on RREF

1. Identify Pivot Columns: In an RREF matrix, each row with a leading 1 (pivot) defines a pivot column. These columns correspond to the variables that are determined uniquely in terms of other variables.
2. Count Non-Pivot Columns: Any column that does not contain a leading 1 (pivot) is a non-pivot column. Non-pivot columns correspond to the free variables of the system.
3. Number of Free Parameters: The number of free parameters is equal to the number of non-pivot columns.

ex $\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ - 2 pivot columns
- 2 non-pivot columns
↳ 2 free parameters

1) Bring back to SoEq:

$\begin{cases} x_1 - x_3 = 0 \\ x_2 + 2x_3 = x_4 \end{cases} \Leftrightarrow \begin{cases} x_1 = x_3 = t \\ x_2 = s - 2t \\ x_3 = t \\ x_4 = s \end{cases}$

2) Represent in parametric form

$\Leftrightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}$

So, 2 parameters, based on the 2 non-pivot columns x_3, x_4

VECTORS AND MATRICES

Vectors

Vector: An ordered list of numbers, often representing points or directions in space. For example, a vector R^n in has n components.

$$\underline{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Magnitude (Length): The length of a vector v is given by:

$$\|\underline{v}\| = \sqrt{v_1^2 + v_2^2 + \dots v_n^2}$$

Unit Vector: A vector of length 1, typically representing direction. To find a unit vector $\hat{v}$ in the same direction as v:

$$\hat{v} = \frac{v}{\|v\|}$$

Vector Addition:

$$\underline{u} + \underline{v} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

Vector Scalar Multiplication

$$c \cdot \underline{v} = c \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}$$

Solving a matrix equation

$$\underline{\underline{A}} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \quad \underline{\underline{B}} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \qquad \underline{\underline{A}}\, x = \underline{\underline{B}}$$

1) Write down unknown as vector / matrix

$$x = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$$

2) Set up equation for x

$$\underline{\underline{x}} = \underline{\underline{A}}^{-1} \cdot \underline{\underline{B}}$$

3) Find $\underline{\underline{A}}^{-1}$

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array}\right] \leftrightarrow \left[\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -2 & 1 \end{array}\right] \qquad \underline{\underline{A}}^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$$

4) Find $\underline{\underline{x}}$

$$\underline{\underline{A}}^{-1} \times \underline{\underline{B}} = \begin{array}{cc|cc} & & 1 & 2 \\ & & 1 & 0 \\ \hline 3 & -1 & 2 & 6 \\ -2 & 1 & -1 & -4 \end{array} \qquad \underline{\underline{x}} = \begin{bmatrix} 2 & 6 \\ -1 & -4 \end{bmatrix}$$

1) Express equation with x as a matrix.

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \times \begin{bmatrix} x_1 & x_3 \\ x_2 & x_4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

2) Matrix Multiplication

$$\begin{bmatrix} x_1 + x_2 & x_3 + x_4 \\ 2x_1 + 3x_2 & 2x_3 + 3x_4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

3) Rref of an augmented matrix

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 0 \end{array}\right] \longleftrightarrow \left[\begin{array}{cc|cc} 1 & 0 & 2 & 6 \\ 0 & 1 & -1 & -4 \end{array}\right]$$

$x_1 = 2 \quad x_2 = -1 \quad x_3 = 6 \quad x_4 = -4$

Inconsistent System:

- No solution exists if there is a row where all variable coefficents are zero, but the constant is non-zero

Matrices

Matrix: An array of numbers with m rows and n columns. A matrix $R^{m \times n}$ can be written as:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}$$

m × n – column number

row number

Addition of Matrices

$\underline{\underline{A}}, \underline{\underline{B}} \in \mathbb{F}^{m \times n}$

$$\underline{\underline{A}} + \underline{\underline{B}} = \begin{bmatrix} a_{m} & \cdots & a_{n} \\ \vdots & \ddots & \vdots \\ a_{mn} & \cdots & a_{nn} \end{bmatrix} + \begin{bmatrix} b_{m} & \cdots & b_{n} \\ \vdots & \ddots & \vdots \\ b_{mn} & \cdots & b_{m} \end{bmatrix} = \begin{bmatrix} a_m + b_m & \cdots & a_n + b_n \\ \vdots & \ddots & \vdots \\ a_{mn} + b_{mn} & \cdots & a_{nn} + b_{mn} \end{bmatrix}$$

we can NOT add two matrices that are NOT $m_1 = m_2$, $n_1 = n_2$

Scalar Mutiplication of Matrices

$\underline{\underline{A}} \in \mathbb{F}^{m \times n}$

$$k \cdot \underline{\underline{A}} = \begin{bmatrix} ka_m & \cdots & ka_n \\ \vdots & \ddots & \vdots \\ ka_{mn} & \cdots & ka_{nn} \end{bmatrix}$$

no requirements, any k can be multiplied

Matrix - Matrix Multiplication

$$\underline{\underline{A}} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 3 \\ 2 & 1 & 3 \end{bmatrix} \qquad \underline{\underline{B}} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\underline{\underline{A}} \cdot \underline{\underline{B}} = \begin{array}{ccc|cc} & & & 1 & 1 \\ & & & 1 & 0 \\ & & & 1 & 2 \\ \hline 1 & 2 & 3 & 6 & 7 \\ 1 & 3 & 3 & 7 & 7 \\ 2 & 1 & 3 & 6 & 8 \end{array}$$

$1\times1 + 2\times1 + 3\times1 = 6$

$1\times1 + 1\times0 + 3\times2 = 7$

$1\times1 + 3\times1 + 3\times1 = 7$

$1\times1 + 0\times1 + 3\times2 = 7$

$2\times1 + 1\times1 + 3\times1 = 6$

$2\times1 + 1\times0 + 3\times2 = 8$

$$\underline{\underline{A}} \cdot \underline{\underline{B}} = \begin{bmatrix} 6 & 7 \\ 7 & 7 \\ 6 & 8 \end{bmatrix}$$

→ $A \in \mathbb{F}^{w \times n}$, $B \in \mathbb{F}^{n \times s}$, n has to be the same #

→ Matrix Multiplication is NOT commutative

Dependent vs independent

Dependent

- At least 1 free variable
  - ↳ 1 ≥ columns w/o pivot in rref
- sltn expressed as free parameters
- det $(\underline{\underline{A}}) = 0$
- Not a full rank
- Free-parameter / zero-row

Independent

- system has no free variables (each variable is a leading variable in a pivot column)
- each variable has a unique sltn with no free parameters
- Full rank
- If all answers are 0

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

- det $(\underline{\underline{A}}) \neq 0$

Square Matrices

Square Matrix: Matrix with the same number of rows & columns

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \in \mathbb{F}^{n \times n}$$

$$\underbrace{\begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{nn} \end{bmatrix}}_{\text{Diagonal}} \quad \underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}}_{\text{Upper Triangular}} \quad \underbrace{\begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}}_{\text{lower Triangular}}$$

Identity Matrix:

$$\underline{\underline{I}}_n = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \in \mathbb{F}^{n \times n}$$

Transpose matrix:

$$\underline{\underline{A}} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \qquad \underline{\underline{A}}^T = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

Inverse Matrix

- has an inverse if square & det $(A) \neq 0$

1) Augmented Matrix with identity matrix on the right:

$$\underline{\underline{A}} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \qquad [\underline{\underline{A}} \mid \underline{\underline{I}}_2] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array}\right]$$

2) Row Reduction until identity matrix on the left:

$$= \left[\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -2 & 1 \end{array}\right]$$

3) The inverse is now on the right.

$$\underline{\underline{A}}^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$$

1) Find determinant

$$\begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix} \quad \underbrace{(2 \cdot 1 \cdot 3)}_{6} + \underbrace{(0 \cdot 0 \cdot 0)}_{0} + \underbrace{(5 \cdot 1 \cdot -1)}_{-5} - \big(\underbrace{(0 \cdot 1 \cdot -1)}_{0} + \underbrace{(5 \cdot 0 \cdot 3)}_{0} + \underbrace{(2 \cdot 0 \cdot 1)}_{0}\big)$$

$6 - 5 = 1$

2) $A^{-1} = \frac{1}{|A|} \cdot (\text{adj}(A))^t$

3) Find $\frac{1}{|A|} = \frac{1}{1} = 1$

4)
$$\text{adj}(A) = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -15 & 5 \\ -1 & 6 & -2 \\ 1 & -5 & 2 \end{bmatrix}$$

5)
$$(\text{adj}(A))^t = \begin{bmatrix} 3 & 1 & 1 \\ 15 & 6 & 5 \\ 5 & 2 & 2 \end{bmatrix}$$

6)
$$A^{-1} = 1 \cdot \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$$

DETERMINANTS

Determinant of a square matrix

Determinant: A scalar value that can be computed from the elements of a square matrix. It provides important information about the matrix, such as whether it is invertible.

↳ For an nxn matrix, determinant is expressed as $\det(A)$ or $|A|$

2x2 Matrix Determinant

$\underline{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\quad |\underline{A}| = ad - bc$

Properties

1) Invertibility: invertible if $\det(\underline{A}) \neq 0$
2) Multiplicative Property:

$\det(AB) = \det(A) \cdot \det(B)$

3x3 Matrix Determinant

1) Choose column / row with most zeros.

ex

$\underline{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ $\quad |\underline{A}| = a(ei-hf) - b(di-gf) + c(dh-ge)$

2) Remember the following signs

$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

$$\det(\underline{A}) = \begin{cases} a_{11} & \text{if } n=1 \\ \sum_{i=1 \text{ or } j=1}^{n} (-1)^{i+j} a_{ij} \underbrace{\det(\overbrace{\underline{A}(i,j)}^{\text{Submatrix}})}_{\text{minor}} & \text{if } n>1 \end{cases}$$

Submatrix

$\underline{A} \in \mathbb{F}^{n \times n}$

$\underline{A}(i,j)$ — Row Column — Matrix where row i and column j are removed

Determinant of a triangular matrix:

$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & \ddots & 0 \\ a_{n1} & a_{n2} & \vdots & a_{nn} \end{bmatrix}$ ← Triangular Matrix

1st Row or 1st Column will always have the most 0s

$$\boxed{\begin{aligned} \det(\underline{A}) &= a_{11} \det(\underline{A}(1,1)) \\ &= a_{11} \cdot a_{22} \cdot a_{33} \cdot a_{44} \cdots a_{nn} \end{aligned}}$$

Determinants and elementary row operations

Effect of elementary row operations on the determinant:

1) Row Swapping (↕): Swapping 2 rows of a matrix changes the sign of the determinant. If $\det(\underline{A}) = d$, $\det(\underline{A})$ where $\underline{A}$ swapped rows is $-d$
2) Row Scaling $(k \cdot R_i)$: Multiplying a row by scalar k, scales the determinant by k. $\det(\underline{A}) = d$
   ↳ $\det(\underline{A}) = kd$
3) Inverse of matrix $(\underline{A}^{-1})$: Inverting the matrix, inverts the determinant
   ↳ $\det(\underline{A}) = d$
   $\det(\underline{A}^{-1}) = \frac{1}{d}$
4) Transverse of matrix $(\underline{A}^T)$: The transverse of a matrix $\underline{A}$, $\underline{A}^T$ will have the same determinant
   $\det(\underline{A}) = \det(\underline{A}^T)$
5) Row Replacement $(R_i + kR_j)$: Adding a multiple of one row to another will NOT change the determinant
6) $\det(\underline{A} \cdot \underline{B}) = \det(\underline{A}) \cdot \det(\underline{B})$

VECTOR SPACES

Definition and examples of vector spaces

Vector Space:

→ is a set V over a field $\mathbb{F}$ where we define 2 operations:

- addition
- scalar multiplication

such that: V is stable / closed; and these are fulfilled:

(i) $u+(v+w)=(u+v)+w$

(ii) $u+v=v+u$

(iii) $u+0=u$ zero vector

(iv) $u+(-u)=0$

(v) $c\cdot(d\cdot u)=(c\cdot d)\cdot u$

(vi) $1\cdot u=u$

(vii) $c\cdot(u+v)=c\cdot u+c\cdot v$

(viii) $(c+d)\cdot u=c\cdot u+d\cdot u$

Stability:

→ Vector spaces must be stable under both addition and scalar multiplication.

↳ Either of those operations yields a result that remains in the same set.

$\underline{a}+k\underline{b}\in V$

Proving something is a vector space:

In general:
$\mathbb{R}^N$ over $\mathbb{R}$
$\mathbb{C}^N$ over $\mathbb{C}$
$\mathbb{C}^N$ over $\mathbb{R}$
$\Bigg\}$ vector spaces

ex Is $\mathbb{R}^3$ over $\mathbb{R}$ a vector space?

1) Prove stability

1.1) Choose representative vectors

$$\underline{a}=\begin{bmatrix}a_1\\a_2\\a_3\end{bmatrix},\quad \underline{b}=\begin{bmatrix}b_1\\b_2\\b_3\end{bmatrix},\quad k\in\mathbb{R}$$

1.2) check using $\underline{a}+k\underline{b}$

$$\underline{a}+k\underline{b}=\begin{bmatrix}a_1\\a_2\\a_3\end{bmatrix}+\begin{bmatrix}b_1k\\b_2k\\b_3k\end{bmatrix}=\begin{bmatrix}a_1+b_1k\\a_2+b_2k\\a_3+b_3k\end{bmatrix}$$

Still a $\mathbb{R}^3$, so it is stable

2) Prove properties

2.1) Example: $\underline{a}+\underline{b}=\begin{bmatrix}a_1+b_1\\a_2+b_2\\a_3+b_3\end{bmatrix}=\begin{bmatrix}b_1+a_1\\b_2+a_2\\b_3+a_3\end{bmatrix}=\underline{b}+\underline{a}$

NOTE: $\mathbb{R}^3$ over $\mathbb{C}$ would NOT be a vector space as $\begin{bmatrix}1\\2\\3\end{bmatrix}\cdot i\notin\mathbb{R}^3$ $\quad k\in\mathbb{C}$

Checking polynomials for vector space:

EX $\{az^2+bz+c \mid a,b,c\in\mathbb{R}\}$ over $\mathbb{R}$

1) Stability: Choose 2 polynomials p & z

$$p(z)+k\cdot q(z)=(p_1z^2+p_2z+p_3)+k(q_1z^2+q_2z+q_3)$$
$$=(p_1+kq_1)z^2+(p_2+kq_2)z+(p_3+kq_3)$$

Still a 2nd degree polynomial, therefore, stable

2) Properties

Choosing a basis for polynomial vector space

1) come up w/ linearly indep. polynomials as hypothetic basis

$$\beta=(\underbrace{1+z+z^2}_{a},\ \underbrace{2+3z}_{b},\ \underbrace{1}_{c})$$

keep adjusting for descending degrees step by step

2) $L=5+3z+4z^2=\frac{3}{4}a+\frac{1}{3}b+3c \iff [L]_\beta=\left(\frac{3}{4},\frac{1}{3},3\right)$

Basis of a vector space

Basis: set of vectors in a vector space that is:

1) Linearly Independent

↳ No vector in basis set can be written as a linear combination of others

2) Span → Any vector V can be written as a linear combination of basis vectors

Dimension: number of basis vectors

ex Finding and proving basis:

$\mathbb{R}^3$ basis is $v_1=\begin{bmatrix}1\\0\\0\end{bmatrix}$ $v_2=\begin{bmatrix}0\\1\\0\end{bmatrix}$ $v_3=\begin{bmatrix}0\\0\\1\end{bmatrix}$

1) Prove linear independence

$$c_1v_1+c_2v_2+c_3v_3=0$$

$$c_1\begin{bmatrix}1\\0\\0\end{bmatrix}+c_2\begin{bmatrix}0\\1\\0\end{bmatrix}+c_3\begin{bmatrix}0\\0\\1\end{bmatrix}=0 \iff \begin{cases}c_1=0\\c_2=0\\c_3=0\end{cases}\to \text{Lin. Indep.}$$

2) Prove spanning

$$v=\begin{bmatrix}x\\y\\z\end{bmatrix}\qquad v=c_1v_1+c_2v_2+c_3v_3 \iff \begin{cases}c_1=x\\c_2=y\\c_3=z\end{cases}$$

Since x, y, z can be chosen freely, we can always find $c_{1,2,3}$ which will fit these equations.

Ordered-basis

↳ A basis where the order matters, as it defines a coordinate system. So, changing the order of basis vectors changes the coordinates of vectors in that basis.

ex An ordered basis for $\mathbb{R}^2$ could be (v_1,v_2) $\quad v_1=\begin{bmatrix}1\\0\end{bmatrix}$ $v_2=\begin{bmatrix}0\\1\end{bmatrix}$

↳ For any vectors $x\in\mathbb{R}^2$ $\quad x=x_1v_1+x_2v_2$

If we changed the order to (v_2,v_1) the coordinates would instead be (x_2,x_1)

Checking matrix in a vector space

Set of matrices $\mathbb{R}^{2\times2}$ $\begin{bmatrix}1&2\\3&0\end{bmatrix},\begin{bmatrix}0&0\\1&4\end{bmatrix}$ vector space?

1) Stability: is $\underline{\underline{A}}+k\underline{\underline{B}}\in\mathbb{R}^{2\times2}$

$$\begin{bmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{bmatrix}+k\begin{bmatrix}b_{11}&b_{12}\\b_{21}&b_{22}\end{bmatrix}\longleftrightarrow\begin{bmatrix}a_{11}+kb_{11}&a_{12}+kb_{12}\\a_{21}+kb_{21}&a_{22}+kb_{22}\end{bmatrix}\in\mathbb{R}^2$$

↑ So, stable

2) Check Properties

Choosing basis for matrix vector space:

$$\mathcal{E}=\left(\begin{bmatrix}1&0\\0&0\end{bmatrix},\begin{bmatrix}0&1\\0&0\end{bmatrix},\begin{bmatrix}0&0\\1&0\end{bmatrix},\begin{bmatrix}0&0\\0&1\end{bmatrix}\right)\qquad \dim_{\mathcal{E}}[V]=4$$

ex $\begin{bmatrix}1&2\\3&0\end{bmatrix}_{\mathcal{E}}=(1,2,3,0)$

Subspaces of a vector space

Subspace: subset $W\subset V$ that is also a vector space under the same addition & scalar multiplication defined on V.

1) zero vector of W is in V

2) if $u,v\in W$, $k\in F$ $\quad \underline{u+ku\in W}$

ex 1) In $\mathbb{R}^2$, any line through origin is a subspace

2) soln space of homogeneous system. Set of all solutions $Ax=0$ forms a subspace

Properties:

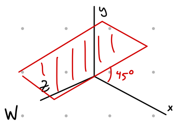
1) Any subspace of a finite-dimensional vector space is itself finite-dimensional

2) intersection of 2 subspaces is a subspace

Prove a subspace:

- we only have to check for stability in the vector space, as all properties are inherited from parent vector space.

$$W=\left\{\begin{bmatrix}a\\b\\c\end{bmatrix}\ \middle|\ a,c\in\mathbb{R}\right\}$$



$$\underline{a}+k\underline{b}=\begin{bmatrix}a_1\\a_1\\a_3\end{bmatrix}+k\begin{bmatrix}b_1\\b_1\\b_3\end{bmatrix}=\begin{bmatrix}a_1+kb_1\\a_1+kb_1\\a_3+kb_3\end{bmatrix}\in W$$

$a_1\ a_1\ b_1\ b_1$ because for a 45 angle, two coordinates should be the same (like the slope of $y=x$)

Disprove a subspace by counter-example:

$$Q=\left\{\begin{bmatrix}a\\b\\c\end{bmatrix}\in\mathbb{R}^3\ \middle|\ a\cdot b\cdot c=0\right\}$$

1) Define 2 vectors $\begin{bmatrix}1\\2\\0\end{bmatrix}\in Q$ $\quad\begin{bmatrix}0\\3\\7\end{bmatrix}\in Q$

2) Check addition stability: $\begin{bmatrix}1\\2\\0\end{bmatrix}+\begin{bmatrix}0\\3\\7\end{bmatrix}=\begin{bmatrix}1\\5\\7\end{bmatrix}$

$1\cdot5\cdot7\neq0$, therefore $\notin W$ and not stable: Not a vector space

EX Express $\underline{W}$ originally in space e, now in space γ.

1) Define the two ordered basis.

$[\underline{W}]_e=\begin{bmatrix}1\\2\\3\end{bmatrix}$ $\quad e=(\underline{e}_1,\underline{e}_2,\underline{e}_3)$ $\quad \gamma=(v_1,v_2,v_3)$

2) Set up an expression for $[\underline{W}]_e$ is represented by parameters of γ.

$$[\underline{W}]_\gamma=k_1v_1+k_2v_2+k_3v_3$$

$$\begin{bmatrix}1\\2\\3\end{bmatrix}=k_1\begin{bmatrix}1\\1\\1\end{bmatrix}+k_2\begin{bmatrix}2\\2\\3\end{bmatrix}+k_3\begin{bmatrix}4\\3\\1\end{bmatrix}$$

Taken from $v_1\ v_2\ v_3$ from previous examples

3) Use rref, with an augmented matrix including $[\underline{W}]_e$

$$\begin{bmatrix}1&2&4&1\\1&2&3&2\\1&3&1&3\end{bmatrix}\longleftrightarrow\begin{bmatrix}1&0&0&7\\0&1&0&-1\\0&0&1&-1\end{bmatrix}$$

so, $k_1=7$, $k_2=-1$, $k_3=-1$

4) Write down final coordinates

$$[\underline{W}]_\gamma=(7,-1,-1)$$

LINEAR MAPS

Linear maps using matrices

Linear Map: transformation between vector spaces V & W over the same field $\mathbb{F}$, $T: V \to W$, which satisfies
{ 1) Additivity
2) Scalar multiplication }

Linearity: $T(u+cv) = T(u) + cT(v)$

Matrix Representation of a linear map:

If V and W are finite-dimension vector spaces, we can represent linear map $T: V \to W$ as matrix $\underline{A}$, such that for any $v \in V$ $T(v) = Av$

→ columns of A are images of the basis vectors of V under T, expressed as basis of W

Checking Linearity of a map:

① $L: \mathbb{R}^3 \to \mathbb{R}^2$ is given by $T(x,y,z) = (x+y, y+z)$

1) check additivity

1.1) Calculate $T(u+v)$

$u+v = (\overbrace{x_1+x_2}^{x}, \overbrace{y_1+y_2}^{y}, \overbrace{z_1+z_2}^{z})$

$T(u+v) = ((x_1+x_2)+(y_1+y_2), (y_1+y_2)+(z_1+z_2))$

1.2) Simplify $T(u+v)$

$T(u+v) = (x_1+x_2+y_1+y_2, y_1+y_2+z_1+z_2)$

1.3) Calculate $T(u)$, $T(v)$

$T(u) = (x_1+y_1, y_1+z_1)$ $T(v) = (x_2+y_2, y_2+z_2)$

1.4) Find $T(u)+T(v)$

$T(u)+T(v) = (x_1+y_1+x_2+y_2, y_1+z_1+y_2+z_2)$

2) Check scalar multiplication

2.1) Calculate $T(c\underline{v})$

$c\underline{v} = (\overset{x}{cx}, \overset{y}{cy}, \overset{z}{cz})$

$T(c\underline{v}) = (cx+cy, cy+cz)$

2.2) Calculate $c(T(\underline{v}))$

$T(\underline{v}) = (x+y, y+z)$

$cT(\underline{v}) = (c(x+y), c(y+z))$

$= (cx+cy, cy+cz)$

$T(u+v) = T(u)+T(v)$; $T(cv) = cT(v)$ linear

Injectivity & Surjectivity of Functions

Theoretically:

1) L is injective if $\ker L = \{0\}$

$=$ rank $[A]$ = dim $[V_1]$

2) L is surjective if $\operatorname{Im} L = V_2$

$=$ rank $[A]$ = dim $[V_2]$

Linear maps between general vector spaces

Properties of Maps:

1) **Kernel:** Set of vectors in V that map to zero vector in W

$\ker(T) = \{v \in V \mid T(v) = 0\}$ (in polynomials, kernel → root)

2) **Image** (Range): set of vectors in W that are images of vectors in V $\operatorname{im}(T) = \{T(v) \mid v \in V\}$. The image is a subspace of W

• Kernel is a vector subspace if the map is linear.

For $\underline{a}, \underline{b} \in \ker L$, $k \in F$ $L(\underline{a}+k\underline{b}) = L(\underline{a}) + kL(\underline{b})$
$0 + k0$ 0 as part of kernel

• stable, hence subspace

Mapping Matrix: is a matrix that represents a linear transformation T:V→W between vector spaces V and Wwhen specific bases are chosen for these spaces. This matrix allows us to compute the output of the transformation using matrix-vector multiplication.

Finding the Mapping Matrix:

let $T: \mathbb{R}^3 \to \mathbb{R}^2$ be defined as $T(x,y,z) = (x+y, y+z)$

Basis $V = v_1 = \begin{bmatrix}1\\0\\0\end{bmatrix} v_2 = \begin{bmatrix}0\\1\\0\end{bmatrix} v_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$ Basis $W = \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$

1) Choose Bases

2) Apply the linear map to each basis vector of V: $T(v_1) = T(1,0,0) = (1,0)$
$T(v_2) = (1,1)$ $T(v_3) = (0,1)$

3) Express $T(v_j)$ in terms of the basis of W

$T(v_1) = (1,0) = \begin{bmatrix}1\\0\end{bmatrix} = 1 \cdot w_1 + 0w_2$

$T(v_2) = (1,1) = \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix} = 1w_1 + 1w_2$

$T(v_3) = (0,1) = \begin{bmatrix}0\\1\end{bmatrix} = 0w_1 + 1w_2$

4) Construct matrix

$[T] = \begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}$

Finding Image using a mapping matrix:

let $\mathbb{R}^3 \to \mathbb{R}^2$ have mapping matrix $[T] = \begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}$

1) Extract columns of $[T]$

$c_1 = \begin{bmatrix}1\\0\end{bmatrix}$ $c_2 = \begin{bmatrix}1\\1\end{bmatrix}$ $c_3 = \begin{bmatrix}0\\1\end{bmatrix}$

The image of T is span $\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\}$

2) Check linearly independent & identify pivot column

$\begin{bmatrix}1&1&0\\0&1&1\end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix}1&0&-1\\0&1&1\end{bmatrix}$ pivot columns in this case, correspond to c_1 & c_2

3) Find basis

$\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$ The basis is the pivot columns of image

4) Find dimension

$\dim(\operatorname{im}(T)) = 2$ # of pivot columns

Finding kernel using a mapping matrix

1) Solve $[T]x = 0$

$[T]x = \begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$

1.1) Turn into a SoEq

$\begin{cases}x_1+x_2=0\\x_2+x_3=0\end{cases}$

1.2) Define a free variable t

$\begin{cases}x_1=-t\\x_2=t\\x_3=-t\end{cases}$

2) Find parametric form using t

$\ker(T) = $ span $\{t\begin{bmatrix}-1\\1\\-1\end{bmatrix}\}$

Basis of kernel: ⟶ $\begin{bmatrix}-1\\1\\-1\end{bmatrix}$

Rank - Nullity Theorem

for $T: V \to W$

$\dim(V) = \operatorname{rank}(T) + \operatorname{nullity}(T)$

$= \dim(\operatorname{im}(T)) + \dim(\ker(T))$

Using the Rank-Nullity Theorem:

1) Find Mapping Matrix & its Rank

$[T] = \begin{bmatrix}1&1&0\\0&1&0\end{bmatrix}$ $\dim([T]) = \operatorname{rank}[T] = 2$

2) Find nullity (dimension of kernel)

2.1) Find kernel

$[T]x = 0$ $\begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$ $\begin{cases}x+y=0\\y+z=0\end{cases}$

$\begin{bmatrix}x\\y\\z\end{bmatrix} = t\begin{bmatrix}-1\\1\\-1\end{bmatrix}$ Span $\{\begin{bmatrix}-1\\1\\-1\end{bmatrix}\}$

1 column ⟶ dimension = 1

Nullity = 1

3) Verify Theorem

Since $V = \mathbb{R}^3$ $\dim(V) = 3$

$\dim(V) = \operatorname{rank}(T) + \operatorname{nullity}(T)$

$3 = 2 + 1$ ✓

Change of Basis Problems:

General Method:

1) Determine the transition matrix $P_{\beta \to \gamma}$

↳ Columns of $P_{\beta\to\gamma}$ are coordinates of basis vectors of β expressed in γ.

2) Convert the vector v from β to γ.

↳ If $[V]_\beta$ is a coordinate vector in V in basis β then $[V]_\gamma = P_{\beta\to\gamma} \cdot [V]_\beta$

3) Convert a vector v from basis γ to β

↳ compute the inverse of $P_{\beta\to\gamma}$

$[V]_\beta = P_{\gamma\to\beta} \cdot [V]_\gamma$

Specific Example if V is a polynomial:

$V = P_2(\mathbb{R})$ polynomials deg ≤ 2

$\beta = \{1, x, x^2\}$ $\gamma = \{1, x-1, (x-1)^2\}$

$V = p(x) = 2+3x+4x^2$

1) Express each element of γ in terms of β

$P_0(x) = 1 = a_1\begin{bmatrix}1\\0\\0\end{bmatrix} + a_2\begin{bmatrix}0\\x\\0\end{bmatrix} + a_3\begin{bmatrix}0\\0\\x^2\end{bmatrix} = a_1 + 0a_2 + 0a_3$
$= \begin{bmatrix}1\\0\\0\end{bmatrix}$

$P_1(x) = x-1 = a_1\begin{bmatrix}1\\0\\0\end{bmatrix} + a_2\begin{bmatrix}0\\x\\0\end{bmatrix} + a_3\begin{bmatrix}0\\0\\x^2\end{bmatrix} = -a_1 + a_2 + 0a_3 = \begin{bmatrix}-1\\1\\0\end{bmatrix}$

$P_2(x) = (x-1)^2 = x^2-2x+1 = a_1 - 2a_2 + a_3 = \begin{bmatrix}1\\-2\\1\end{bmatrix}$

2) Combine column to create a change of basis matrix

$\begin{bmatrix}1&-1&1\\0&1&-2\\0&0&1\end{bmatrix} = {}_\gamma[P]_\beta$

3) Represent v ($p(x) = 2+3x+4x^2$) in β basis

$[p(x)]_\beta = \begin{bmatrix}2\\3\\4\end{bmatrix}$

3.1) Convert to γ

${}_\gamma[P]_\beta \cdot [V]_\beta = [V]_\gamma$

			2
			3
			4
1	-1	1	3
0	1	-2	-5
0	0	1	4

$[V]_\gamma = \begin{bmatrix}3\\-5\\4\end{bmatrix}$

4) You can also check using an inverse

Specific Example if V is a vector:

$V = \begin{bmatrix}3\\5\end{bmatrix}$ Let $V = \mathbb{R}^2$ with $\beta = \{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\}$, $\gamma = \{\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\}$

1. Express basis vectors of β in terms of γ

$b_1 = \begin{bmatrix}1\\0\end{bmatrix}$ in γ $\begin{bmatrix}1\\0\end{bmatrix} = a_1\begin{bmatrix}1\\1\end{bmatrix} + a_2\begin{bmatrix}0\\1\end{bmatrix}$ $a_1 = -1$, $a_2 = 1$ so $b_1 = -c_1 + c_2$

$b_2 = \begin{bmatrix}1\\1\end{bmatrix}$ in γ $\begin{bmatrix}1\\1\end{bmatrix} = a_1\begin{bmatrix}1\\1\end{bmatrix} + a_2\begin{bmatrix}0\\1\end{bmatrix}$ $a_1 = 0$ $a_2 = 1$ so $b_1 = 0c_1 + c_2$

2. Use the values to create a transition matrix.

$\begin{bmatrix}-1&0\\1&1\end{bmatrix} = P_{\beta\to\gamma}$

3. Convert V from β to γ

3.1 Express V in β as $[V]_\beta$

$\begin{bmatrix}3\\5\end{bmatrix} = x_1\begin{bmatrix}1\\0\end{bmatrix} + x_2\begin{bmatrix}1\\1\end{bmatrix}$ $x_1 = 3$ $x_2 = 2$ $[V]_\beta = \begin{bmatrix}3\\2\end{bmatrix}$

3.2 Convert to γ

${}_\gamma[P]_\beta \cdot [V]_\beta = [V]_\gamma = \begin{bmatrix}-1&0\\1&1\end{bmatrix}\begin{bmatrix}3\\2\end{bmatrix} = \begin{bmatrix}-3\\5\end{bmatrix}$

4. You can check through the inverse of ${}_\gamma[P]_\beta$, ${}_\beta[P]_\gamma$

Finding a general sltn to $A \cdot V = b$

Solve: $A \cdot V = b$ $A = \begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$ $b = \begin{bmatrix}6\\15\\24\end{bmatrix}$

1) Find particular sltn v_p

1.1) Row-Reduce $[A|b]$

$\left[\begin{array}{ccc|c}1&2&3&6\\4&5&6&5\\7&8&9&24\end{array}\right] \longleftrightarrow \left[\begin{array}{ccc|c}1&2&3&6\\0&-3&-6&-9\\0&0&0&0\end{array}\right]$

1.2) Use sltn to find particular sltn

$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}6\\-9\\0\end{bmatrix}$ $V_p = \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}0\\3\\0\end{bmatrix}$

2) Find kernel of A

$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} \longleftrightarrow \left[\begin{array}{ccc|c}1&2&3&0\\0&-3&-6&0\\0&0&0&0\end{array}\right]$

$\begin{cases}x_1+2x_2+3x_3=0\\-3x_2-6x_3=0\end{cases}$ $\begin{cases}x_1 = 4t-3t = t\\x_2 = -2t\\x_3 = t\end{cases}$

$\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = t\begin{bmatrix}1\\-2\\1\end{bmatrix}$ $\ker(A) = $ span $\{\begin{bmatrix}1\\-2\\1\end{bmatrix}\}$

3) Covert to a sltn set

$V = V_p + k(\ker) = \begin{bmatrix}0\\3\\0\end{bmatrix} + t\begin{bmatrix}1\\-2\\1\end{bmatrix} = \begin{bmatrix}t\\3-2t\\t\end{bmatrix}$

The Eigenvalue Problem and Diagonalization

11.1 Eigenvalues and Eigenvectors

Eigenvalue (λ): a scalar such that for a linear map $L: V \to V$ or a matrix A, there exists a non-0 vector v where $L(v)$ or $A \cdot v = \lambda v$

Eigenvector (v): A non-zero vector associated with an eigenvalue that satisfies $L(v) = \lambda v$ or $A \cdot v = \lambda v$

- Eigenvectors represent directions in the vector space which remain scaled (not rotated) under a linear transformation
- Eigenvalues determine the scaling factor along their corresponding vectors
- To find eigenvalues for a square matrix $A \in \mathbb{F}^{n \times n}$ solve this equation:

$$\det(A - \lambda I) = 0$$

Finding eigenvalues

ex

1) Derive the equation

$$Av = \lambda v$$
$$Av = \lambda I v$$
$$Av - \lambda I v = 0$$

2) Solve for determinant

$$|A - \lambda I| = 0$$

3) Plug in known values & identity matrix

$$\left| \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

4) Combine into a single matrix

$$= \begin{vmatrix} -6-\lambda & 3 \\ 4 & 5-\lambda \end{vmatrix} = 0$$

5) Solve for λ using a determinant

$$(-6-\lambda)(5-\lambda) - 3 \times 4 = 0$$
$$\lambda^2 + \lambda - 42 = 0$$
$$\lambda = -7, 6$$

Finding the eigenvector

1) Start with $Av = \lambda v$, and plug in known values

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 6 \begin{bmatrix} x \\ y \end{bmatrix}$$

2) Put into a SoEq

$$\begin{cases} -6x + 3y = 6x \\ 4x + 5y = 6y \end{cases} \to \begin{cases} -12x + 3y = 0 \\ 4x - 1y = 0 \end{cases}$$

$$y = 4x$$

3) Translate obtained value into vector form

$$y = 4x \longrightarrow \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

4) Check

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

OR $(\underline{A} - 6\underline{I})\underline{v} = 0$

1) Form an augmented matrix

$$\Leftrightarrow \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} - 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\Leftrightarrow \left[\begin{array}{cc|c} -12 & 3 & 0 \\ 4 & -1 & 0 \end{array}\right]$$

$$\Leftrightarrow \left[\begin{array}{cc|c} 0 & 0 & 0 \\ 4 & -1 & 0 \end{array}\right]$$

$$\Leftrightarrow \left[\begin{array}{cc|c} 4 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

2) Express through a parameter t

$$4k_1 - k_2 = 0 \qquad k_1 = \tfrac{1}{4}k_2$$
$$4k_1 = k_2 \qquad k_2 = k_2$$

$$\underline{v} = t \begin{bmatrix} \frac{1}{4} \\ 1 \end{bmatrix}, t \in \mathbb{R}$$

$$= t \begin{bmatrix} 1 \\ 4 \end{bmatrix}, t \in \mathbb{R}$$

Systems of Linear Ordinary Differential Equations of Order One with Constant Coefficients

Linear First-Order ODEs

- relates a function of $f(t)$ and $f'(t)$

ex. $f'(t) = a(t)f(t) + q(t)$

where:

$a(t)$ is the coefficient function

$q(t)$ is the forcing functions

anything in this term is already linear, so no need to prove it

Homogeneous vs Inhomogeneous ODEs

Homogeneous: $q(t) = 0$ (no forcing function)

Inhomogeneous: $q(t) \neq 0$

General Solution to a 1st order ODE:

The general solution of a linear first order ODE is:

$$f(t) = e^{P(t)} \int e^{-P(t)} q(t)\, dt + C \cdot e^{P(t)}, \quad \text{where } P(t) = \int a(t)$$

ex $f'(t) = 2f(t) + e^{t}$ — inhomogeneous 1st order DE

1) Find the integral of $a(t)$: $a(t) = 2$ $\quad P(t) = \int 2\, dt = 2t$

2) Find the general solution:

$$f(t) = e^{2t} \underbrace{\int e^{-2t} \cdot e^{t}\, dt}_{\int e^{-t}\, dt \,=\, -e^{-t} + C} + C \cdot e^{2t}$$

$$f(t) = e^{2t}\left(-e^{-t} + C\right) = -e^{t} + Ce^{2t}$$

Converting general complex sltns to real

1) Identify Complex Solutions

$$e^{\lambda t} = e^{(a+bi)t} = e^{at} e^{ibt}$$

Using Euler's Formula: $e^{bit} = \cos(bt) + i\sin(bt)$

$$e^{\lambda t} = e^{at}\left(\cos(bt) + i\sin(bt)\right)$$

2) Combine Conjugate solutions

$$C_1 e^{(a+bi)t} + C_2 e^{(a-bi)t}$$

which is the same as

$$C_1 e^{at}\left(\cos(bt) + i\sin(bt)\right) + C_2 e^{at}\left(\cos(bt) - i\sin(bt)\right)$$

3) Separate Real & Imaginary Parts

$$e^{at}\left[(C_1 + C_2)\cos(bt) + i(C_1 - C_2)\sin(bt)\right]$$

$C_1 + C_2 = A \qquad C_1 - C_2 = iB$

$$\hookrightarrow e^{at}\left[A\cos(bt) + B\sin(bt)\right]$$

4) Write the real solution

$$= e^{at}\left[A\cos(bt) + B\sin(bt)\right]$$

where A & B are determined by initial conditions. This represents the real valued solution corresponding to the complex roots.

Systems of Linear First-Order ODEs with Constant Coefficients

A system of linear ODEs is written as:

$f'(t) = \underline{A}\,f(t) + \underline{q}(t)$, where $f(t) = [f_1(t), f_2(t), \ldots, f_n(t)]^T$ is a vector of unknown functions

A is the $n \times n$ coefficient matrix

$q(t)$ is the vector of forcing function

Homogeneous vs inhomogeneous:

Homogeneous: $q(t) = 0$

Inhomogeneous: $q(t) \neq 0$

For an inhomogeneous system, the solutions can be written as:

$f(t) = f_h(t) + f_p(t)$, where:

$f_h(t)$: sltn to homogeneous system

$f_p(t)$: particular sltn to an inhomogeneous system

Finding Solutions:

1) Homogeneous:

Solve: $f'(t) = \underline{A}\, f(t)$

Use eigenvalues λ and eigenvectors $\underline{v}$ of $\underline{A}$

$$f(t) = c_1 v_1 e^{\lambda_1 t} + \ldots + c_n v_n e^{\lambda_n t}$$

2) Inhomogeneous:

- Find a particular solution $f_p(t)$ using methods like undetermined coefficients or variation of parameters
- combine $f_h(t)$

ex **Homogeneous System**

$$\begin{bmatrix} f_1'(t) \\ f_2'(t) \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix}$$

1) Find eigenvalues:

$$\det(\underline{A} - \lambda \underline{I_n}) = (2-\lambda)(2-\lambda) = 0$$

$$\hookrightarrow (2-\lambda)^2 = 0 \quad \lambda = 2$$

2) Find eigenvectors:

$$(\underline{A} - 2\underline{I})\underline{v} = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \longrightarrow v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

3) General Solution:

$$\underline{f}(t) = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t}$$

Inhomogeneous System ??

$$f'(t) = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} f(t) + \begin{bmatrix} e^{t} \\ 1 \end{bmatrix}$$

1) Solve homogeneous system:

$\lambda_1 = 2, \quad \lambda_2 = 3 \qquad \underline{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \underline{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

General Solution:

$$\underline{f}_h(t) = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{3t}$$

2) Find particular Solution:

Assume $\underline{f}_p(t) = \begin{bmatrix} Ae^{t} \\ B \end{bmatrix}$

$$A = \frac{1}{2} \qquad B = \frac{1}{3}$$